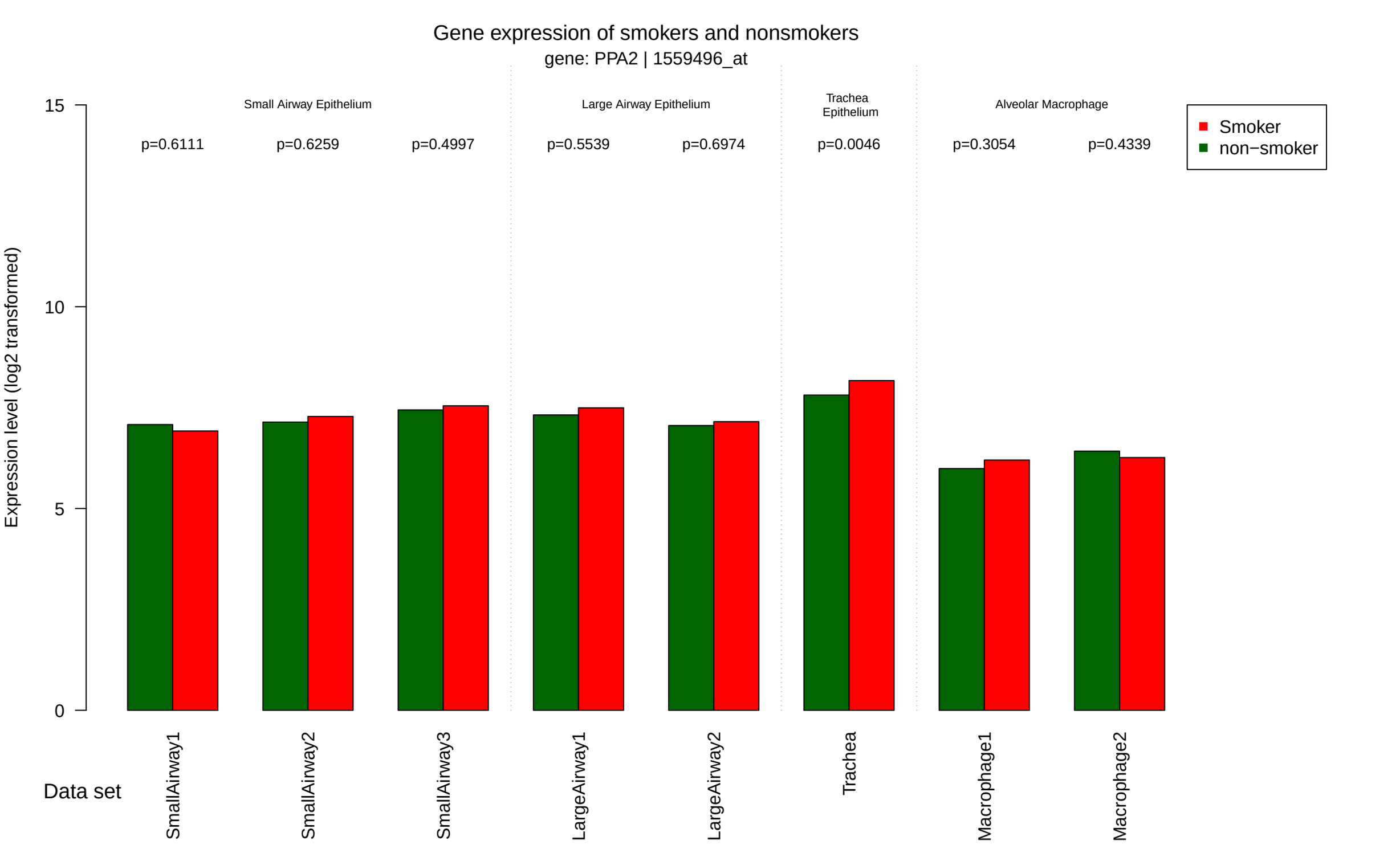
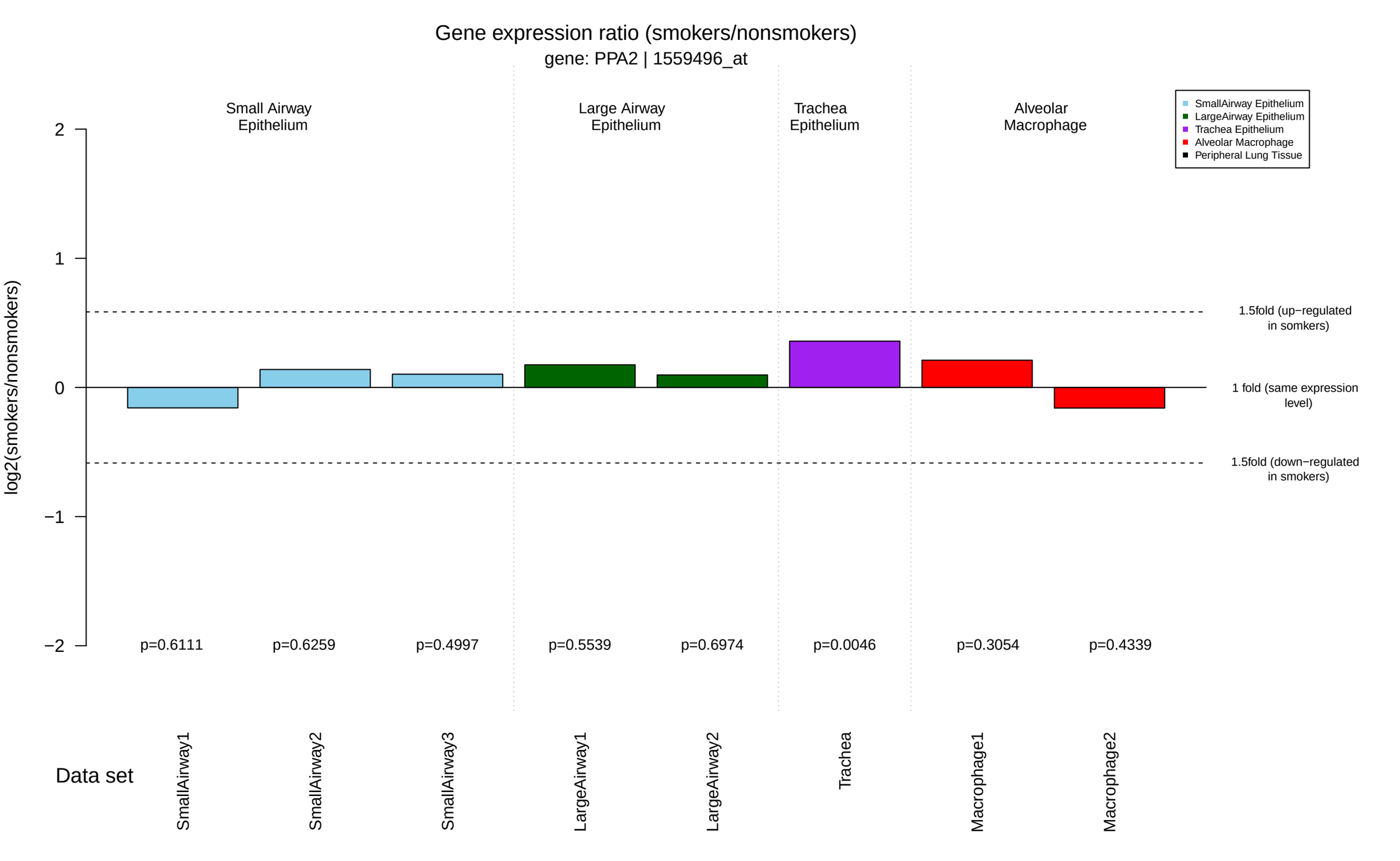
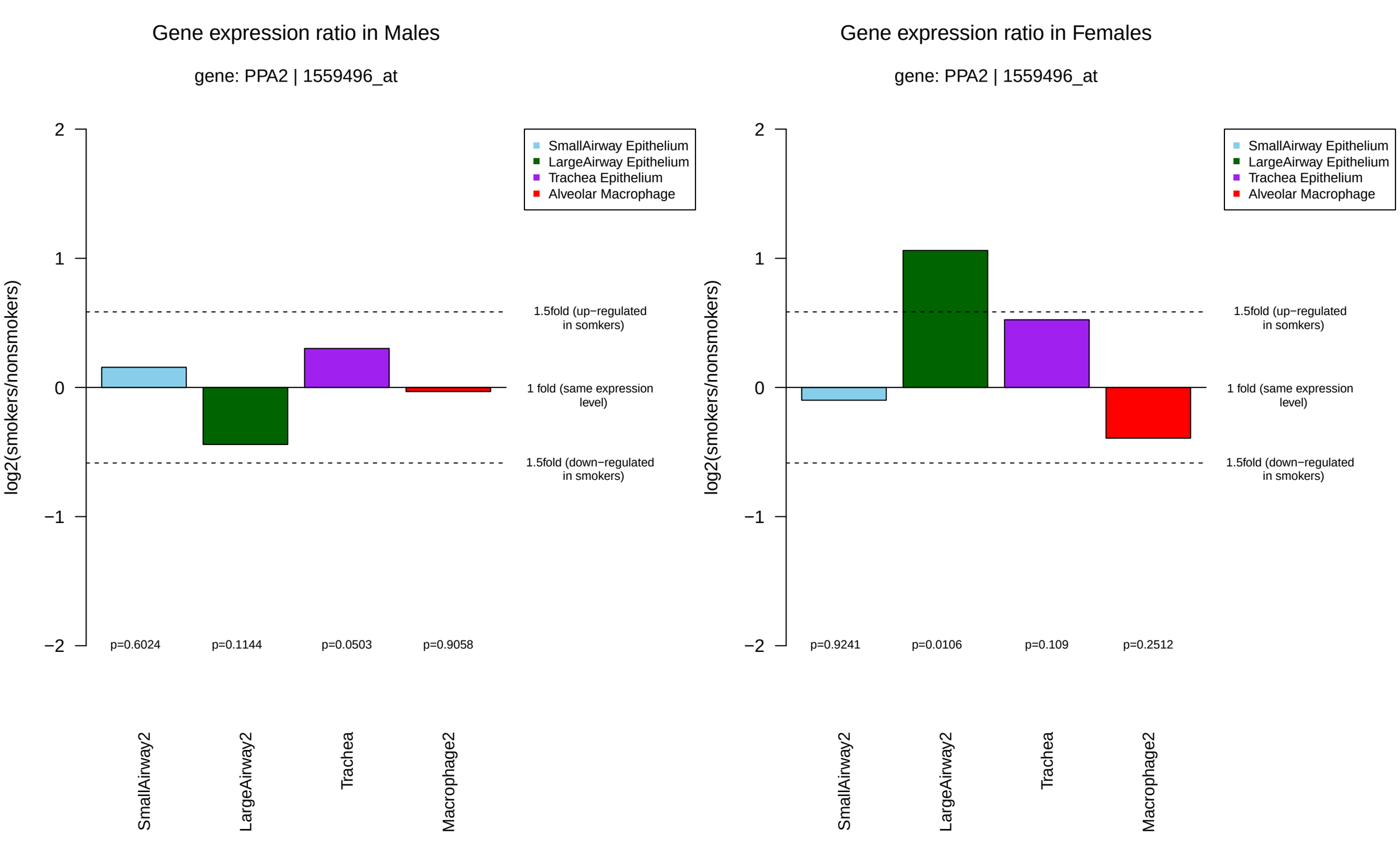
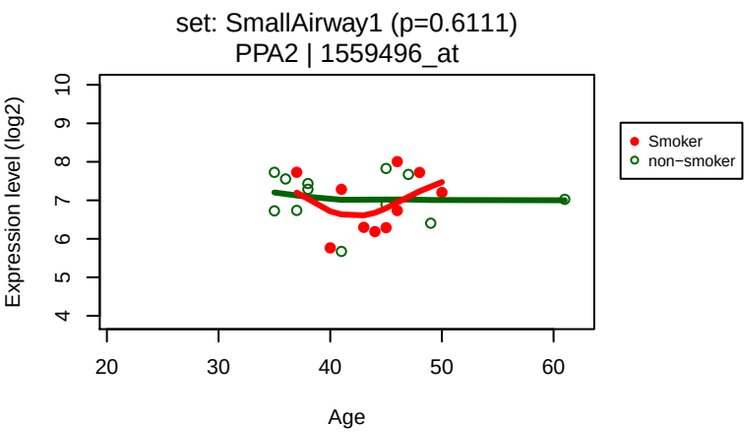
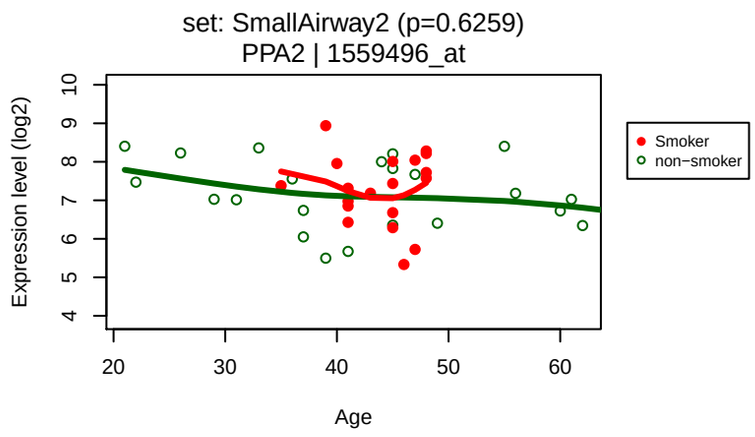
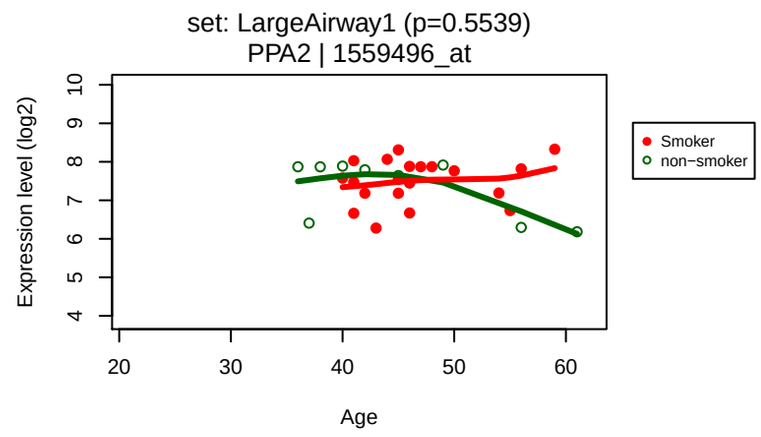

Gene expression of smokers and nonsmokers
gene: PPA2 | 1559496_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6111
p=0.6259
p=0.4997
p=0.5539
p=0.6974
p=0.0046
p=0.3054
p=0.4339
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

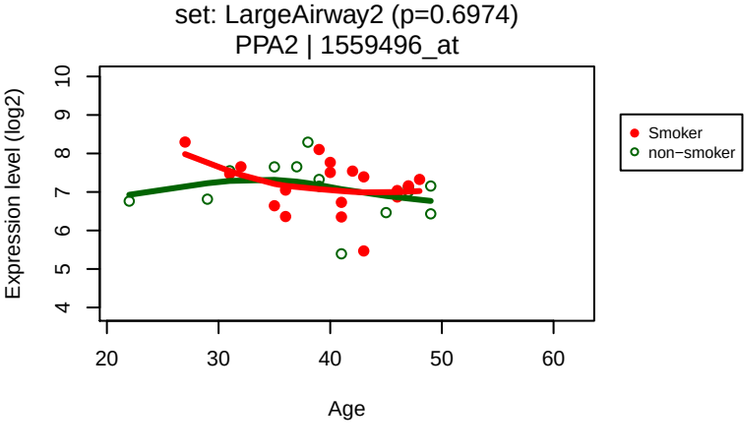

Gene expression ratio (smokers/nonsmokers)
gene: PPA2 | 1559496_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6111
p=0.6259
p=0.4997
p=0.5539
p=0.6974
p=0.0046
p=0.3054
p=0.4339
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

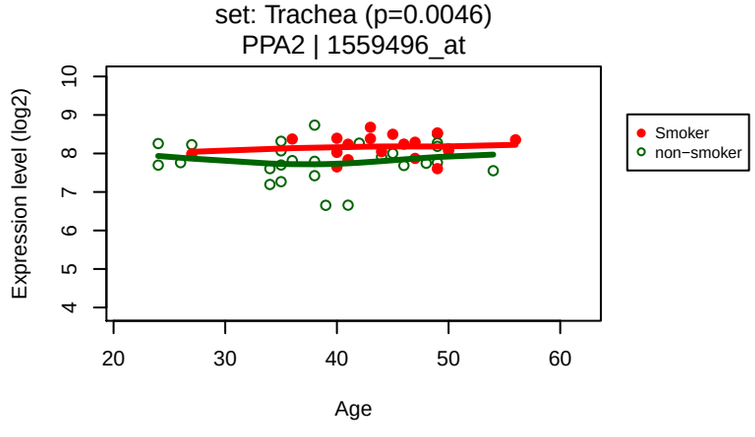

Gene expression ratio in Males
gene: PPA2 | 1559496_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6024
p=0.1144
p=0.0503
p=0.9058
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPA2 | 1559496_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9241
p=0.0106
p=0.109
p=0.2512
SmallAirway2
LargeAirway2
Trachea
Macrophage2

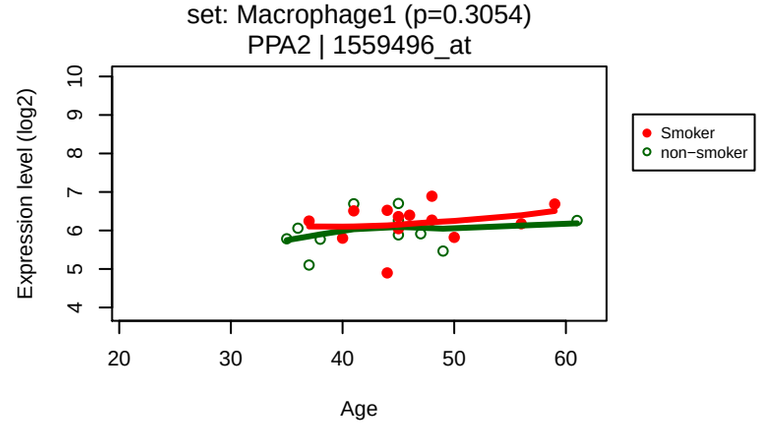

set: SmallAirway1 (p=0.6111)
PPA2 | 1559496_at
Expression level (log2)
Age
Smoker
non−smoker

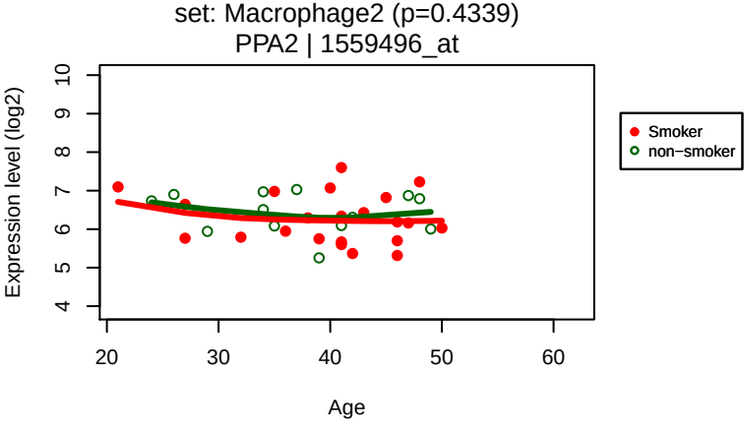

set: SmallAirway2 (p=0.6259)
PPA2 | 1559496_at
Expression level (log2)
Age
Smoker
non−smoker

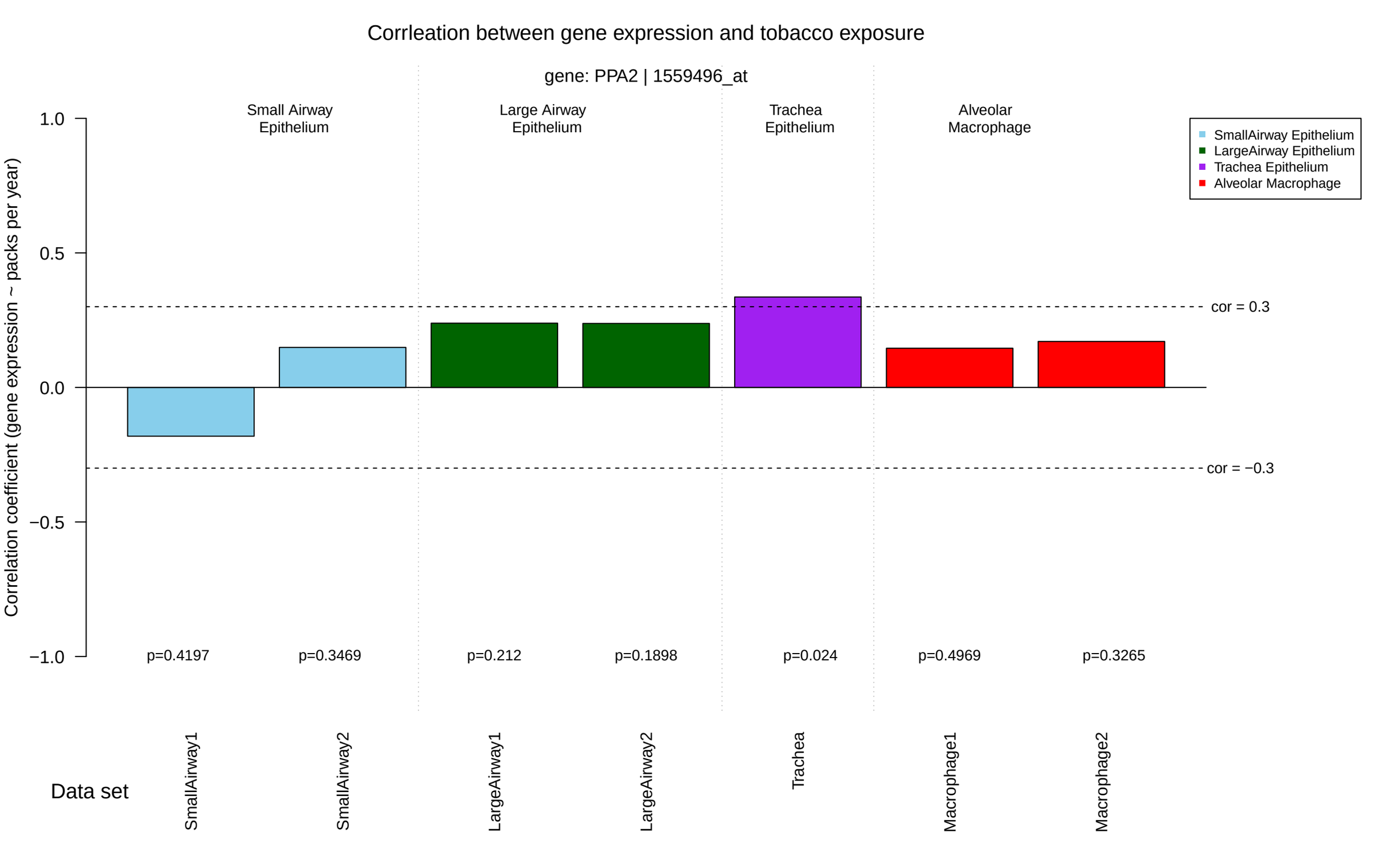

set: LargeAirway1 (p=0.5539)
PPA2 | 1559496_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6974)
PPA2 | 1559496_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0046)
PPA2 | 1559496_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3054)
PPA2 | 1559496_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4339)
PPA2 | 1559496_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPA2 | 1559496_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4197
p=0.3469
p=0.212
p=0.1898
p=0.024
p=0.4969
p=0.3265
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2